

Nirav Patel

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Professional Summary

Software Engineering student with professional experience building and optimizing production backend systems, data pipelines, and full-stack applications using Python, React, and AWS. Combines hands-on engineering experience with systems-focused projects spanning asynchronous workflow orchestration, API development, data infrastructure, and cloud systems.

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks/Libraries: FastAPI, Django, React, Next.js, Node.js, Express

Cloud/DevOps: AWS, Docker, Git, GitHub Actions, CI/CD

Databases/APIs: PostgreSQL, BigQuery, Redis, REST APIs, OpenAI API

Experience

Software Engineering Intern

Apr 2025 - Apr 2026

TrainingHub

Mississauga, ON

- Built and maintained Python REST APIs and React features supporting enrollment and authentication for 500+ learners.
- Developed backend data pipelines for course and session data, shifting analytics updates from weekly to daily.
- Optimized SQL queries (BigQuery) and refined API validation logic, contributing to ~30% faster dashboard load times.
- Collaborated on AWS infrastructure and CI/CD pipelines across the SDLC, supporting reliable production deployments.

Founder and Lead Coding Instructor

Jul 2023 - Present

North Code Academy

Remote

- Led 6 instructors delivering project-based programs to 200+ students with 100% project completion.
- Authored 40+ developer tutorials and code-along projects teaching Web Development with Python and JavaScript.
- Built a Python automation tool integrating REST and OpenAI APIs for rubric-based feedback and progress tracking.

Engineering Outreach Liaison

Feb 2024 - May 2025

Toronto Metropolitan University

Toronto, ON

- Delivered 350+ STEM workshops to 5,000+ students across Ontario, increasing engineering interest by 50%+.
- Built Python, SQL, and Excel pipelines to analyze feedback and improve workshop engagement and operational efficiency.

Projects

AWE | Autonomous Workflow Engine

GitHub

- Architected a FastAPI workflow engine for goal decomposition, DAG orchestration, async execution, and state tracking.
- Built REST/WebSocket interfaces for real-time task-state streaming, failure/retry tracking, and workflow inspection.

Cutr | URL Shortener + Analytics

GitHub

- Engineered a Go/PostgreSQL URL shortener with persistent URL management, redirect lookup, and analytics.
- Built REST endpoints for URL creation, search, analytics, and PIN-protected deletion with a responsive React frontend.

Venture | Full-Stack Music Explorer

GitHub, Demo

- Engineered a Django music explorer with Spotify API integration, auth, response normalization, and debounced search.
- Built a deterministic prompt-to-theme engine mapping natural-language input to curated audio/visuals.

Education

The Pennsylvania State University

GPA: 3.73/4.0

Bachelor of Science, Software Engineering

Expected Graduation: May 2028

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Software Architecture, Database Systems, Networks